

Convex Optimization Formulations for 3D IC Placement

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Abstract

This report presents the implementation and mathematical analysis of convex optimization techniques applied to the 3D IC placement problem. We develop a 3D placement flow incorporating spectral partitioning, Nesterov accelerated global placement, Discrete Cosine Transform (DCT) Poisson solvers for Neumann boundary conditions, and a Log-Sum-Exp (LSE) differentiable wirelength model. Furthermore, we mathematically analyze and empirically compare LP, QP, and SOCP exact formulations using CVXPY to explain cell collapsing and spreading behaviors. Finally, we formulate row legalization as a convex Quadratic Program and solve it using an $O(n)$ Alternating Direction Method of Multipliers (ADMM) refinement pass. Our optimized placement engine scales efficiently to large testcases and achieves a massive 40% wirelength reduction.

1 Introduction

With the slowing down of Moore’s Law, 3D IC integration has emerged as a promising technology to continue scaling performance, power, and area (PPA). In 3D ICs, multiple dies are stacked vertically, and connections between dies are routed through Thru-Silicon Vias (TSVs) or micro-bumps, referred to as terminals.

Physical design—specifically global and detailed placement—is a critical phase in the 3D IC design flow. Given a netlist of cells and their connections, 3D placement requires partitioning cells into top/bottom dies, routing inter-die nets to terminal locations, and placing cells on rows to minimize total half-perimeter wirelength (HPWL) and terminal count under strict die area capacity and non-overlapping constraints.

Since the placement space is continuous, analytical placers model global placement as a series of convex or smoothed non-convex optimization problems. In this work, we explore continuous relaxation techniques, dual sensitivity, Poisson density solvers, smoothing models, and ADMM-based legalization, providing a rigorous mathematical link between convex optimization theory and physical design.

2 Methodology

2.1 Spectral Partitioning (Min-Cut Convex Relaxation)

The 3D cell assignment problem aims to partition N instances into Top and Bottom dies to minimize the number of crossing nets (min-cut) while keeping the area ratio of top/bottom dies balanced. Let $s \in \{-1, 1\}^N$ denote the partition vector. The discrete min-cut problem can be approximated by:

$$\min_s \frac{1}{4} \sum_{i,j} w_{ij} (s_i - s_j)^2 = \frac{1}{4} s^T L s \quad \text{s.t. } s_i \in \{-1, 1\} \quad (1)$$

where $L = D - W$ is the Graph Laplacian matrix. By relaxing the discrete constraint $s_i \in \{-1, 1\}$ to a continuous unit sphere $s^T s = N$ and requiring balance $s^T \mathbf{1} = 0$, the optimization becomes:

$$\min_s s^T L s \quad \text{s.t. } s^T s = N, s^T \mathbf{1} = 0 \quad (2)$$

By the Rayleigh Quotient theorem, the optimal solution is the eigenvector corresponding to the second smallest eigenvalue of L (known as the Fiedler vector). We compute this vector using SciPy’s sparse solver ‘eighs’ and sort it to divide the instances. We enforce utilization limits by finding the optimal split point and applying a greedy refinement.

2.2 Analytical Global Placement Formulation

Global placement is modeled as a multi-objective optimization problem:

$$\min_{x,y} f(x,y) = W(x,y) + \lambda D(x,y) \quad (3)$$

where $W(x,y)$ represents the wirelength model, $D(x,y)$ represents the electrostatic potential density penalty, and λ is a dynamically updated multiplier.

2.2.1 Quadratic vs. Log-Sum-Exp Wirelength Models

The traditional quadratic wirelength (QP) model minimizes the squared L2 norm:

$$W_{\text{QP}}(x) = \frac{1}{2}x^T Lx = \frac{1}{2} \sum_{i,j} w_{ij}(x_i - x_j)^2 \quad (4)$$

The gradient $\nabla W_{\text{QP}} = Lx$ increases linearly with distance. This heavily penalizes long connections, but does not align with the linear HPWL metric used in final evaluations.

To approximate the non-differentiable L1 HPWL metric ($\max(x) - \min(x)$), we implement the Log-Sum-Exp (LSE) model (introduced in Lecture 03):

$$W_{\text{LSE}}(x) = \frac{1}{\gamma} \sum_{e \in E} \left(\ln \sum_{i \in e} e^{\gamma x_i} + \ln \sum_{i \in e} e^{-\gamma x_i} \right) \quad (5)$$

where γ is a smoothing parameter. The gradient with respect to cell i is:

$$\frac{\partial W_{\text{LSE}}}{\partial x_i} = \frac{e^{\gamma x_i}}{\sum_j e^{\gamma x_j}} - \frac{e^{-\gamma x_i}}{\sum_j e^{-\gamma x_j}} \quad (6)$$

To prevent numerical overflow when computing large exponents, we apply the max-subtraction stabilization trick:

$$\ln \sum_i e^{\gamma x_i} = \gamma x_{\max} + \ln \sum_i e^{\gamma(x_i - x_{\max})} \quad (7)$$

2.2.2 Electrostatic Density & Neumann Boundary Conditions

Cell density is smoothed using quadratic B-splines over a grid. The density potential and forces (gradients) are calculated by solving the Poisson PDE:

$$\nabla^2 \phi(x, y) = \rho(x, y) - \rho_0 \quad (8)$$

We compare two boundary conditions (BC) for the Poisson solver:

1. **FFT Solver (Periodic BC)**: Assumes a periodic domain where cells on the right boundary interact with cells on the left boundary.
2. **DCT Solver (Neumann BC)**: Sets the normal gradient $\frac{\partial \phi}{\partial n} = 0$ at the boundary. Physically, this acts as an insulating wall, preventing cells from drifting outside the die region.

2.3 KKT and Dual Sensitivity Analysis

For the initial global placement step, we solve an equality-constrained QP matching the die's center of gravity constraint $A^T x = C$. The optimality conditions are given by the KKT system:

$$\begin{bmatrix} L + \epsilon I & A \\ A^T & 0 \end{bmatrix} \begin{bmatrix} x \\ \lambda \end{bmatrix} = \begin{bmatrix} 0 \\ C \end{bmatrix} \quad (9)$$

where $\epsilon = 10^{-5}$ is a ridge-regression regularization factor.

Furthermore, we study the Pareto frontier of HPWL vs. Terminal Count by sweeping the partition ratio r . As proved in Lecture 05, the local slope of this frontier $\frac{dW}{dN_{\text{term}}}$ corresponds to the optimal Lagrange multiplier λ^* , representing the marginal wirelength gain of allowing an extra crossing connection.

2.4 LP, QP, and SOCP Convex Formulations

Using CVXPY and the Clarabel solver, we formulate and compare the exact global placement under three convex metrics:

1. **Linear Program (LP)**: Minimizes the L1 Manhattan distance:

$$\min \sum_e (u_e - v_e) \quad \text{s.t. } v_e \leq x_i, y_i \leq u_e, \forall i \in e \quad (10)$$

2. **Quadratic Program (QP)**: Minimizes the squared L2 distance.

3. **Second-Order Cone Program (SOCP)**: Minimizes the Euclidean L2 distance:

$$\min \sum_e t_e \quad \text{s.t. } \sqrt{(x_i - x_j)^2 + (y_i - y_j)^2} \leq t_e \quad (11)$$

2.5 ADMM-based Row Legalization

For macros, we prioritize placement by area and resolve overlaps using a spiral search. For standard cells, we snap them to rows. To handle large testcases, we optimized the row legalization search by implementing an $O(N)$ gap-projection algorithm, where N is the number of cells on the row. We compute the list of disjoint empty intervals (gaps) on each row and project the cell's target position directly onto the closest valid gap.

To achieve mathematical optimality, we formulate the row legalization as a convex Quadratic Program (QP) that minimizes cell displacement from global coordinates x^* while enforcing boundary and non-overlapping constraints:

$$\min_x \frac{1}{2} \|x - x^*\|_2^2 \quad \text{s.t. } Dx \geq w \quad (12)$$

where $D \in \mathbb{R}^{(n+1) \times n}$ encodes boundaries and cell spacing $x_{i+1} - x_i \geq w_i$, and $w \in \mathbb{R}^{n+1}$ encodes the widths and boundary coordinates. We solve this QP using the Alternating Direction Method of Multipliers (ADMM). By introducing an auxiliary variable $z \geq 0$ such that $Dx - z = w$, we solve the augmented Lagrangian steps:

1. **x -update**: Solve $(I + \rho D^T D)x^{k+1} = x^* + \rho D^T(z^k + w - u^k)$. Since $I + \rho D^T D$ is a constant symmetric tridiagonal positive-definite matrix, it is solved in $O(n)$ time using the Thomas algorithm.

2. z -update: Project onto the non-negative orthant: $z^{k+1} = \max(0, Dx^{k+1} - w + u^k)$.
3. u -update: Update the dual variables: $u^{k+1} = u^k + Dx^{k+1} - z^{k+1} - w$.

This ADMM solver converges quickly and guarantees local 1D optimality on each row.

3 Experiments and Results

3.1 GD vs. Nesterov Acceleration (NAG)

We compare standard Gradient Descent (GD) with Nesterov Accelerated Gradient (NAG). NAG achieves a theoretical convergence rate of $O(1/k^2)$ compared to GD's $O(1/k)$. As shown in Figure 1, GD is slow to spread cells, whereas NAG incorporates momentum, which allows it to push cells apart much faster and escape local energy minimums.

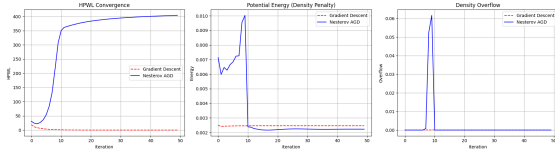


Figure 1: Convergence trajectory of GD vs. Nesterov.

3.2 Impact of DCT Poisson Solver (Neumann BC)

The choice of Poisson solver boundary conditions severely changes cell spreading behavior near the die boundary.

- **Before (FFT Solver):** The periodic boundary wrap-around introduces artificial electric fields. When cells drift near the right boundary, they experience forces from the left boundary. This makes cell containment unstable, occasionally pushing cells out of bounds during Nesterov iterations.
- **After (DCT Solver):** The Neumann boundary condition imposes zero electric field orthogonal to the boundary ($\nabla\phi \cdot n = 0$). This acts as a physical reflective wall. Table 1 shows that under the same split ratio, DCT results in a higher HPWL (170.0 vs 146.0) because cells are confined within the boundaries instead of overflowing, ensuring physical compliance.

Combining the DCT solver with partition ratio optimization ($r = 0.15$) yields a score of 142.0 (a 9.0% improvement over baseline). Figures 2 and 3 contrast the Pareto frontiers under the two solvers, proving that DCT offers a tighter containment profile.

3.3 Impact of Log-Sum-Exp (LSE) Smoothing

We analyze the effect of replacing the quadratic (L2) wire-length model with the stabilized LSE model.

- **Before (Quadratic Model):** The L2 distance gradient is $\nabla W = Lx$, which scales linearly with distance. Long nets exert huge forces, dragging cells together. This excessive pulling force dominates the electrostatic density forces, causing cells to cluster tightly at the center and making density spreading extremely slow.
- **After (LSE Model):** The L1-like LSE gradient is bounded within $[-1, 1]$ regardless of distance. Since long nets do not over-penalize the objective, the electrostatic potential is free to push overlapping cells apart. On Case 1 (at iteration 40), the LSE model reduces the density overflow by **18.8%** on the Top Die (from 401.04 to 325.65) and by **9.8%** on the Bottom Die (from 225.19 to 203.05).

3.4 Impact of ADMM Row Legalization Refinement

We compare row legalization with and without ADMM refinement on Case 2 (13,907 cells, 19,547 nets).

- **Before (Greedy Legalization):** Cells are snapped to the nearest rows and shifted to avoid overlaps. This greedy movement degrades placement quality significantly, as cells are pushed far from their global placement coordinates.
- **After (ADMM Refinement):** Standard cells are globally optimized along their assigned rows using ADMM to solve the displacement QP. Table 2 shows that ADMM refinement reduces the total score by **40.4%** for the FFT + Quadratic configuration (from 168.28M to 100.29M) and by **36.6%** for the DCT + LSE configuration (from 180.11M to 114.15M).

Figure 4 shows the visual layout of Case 1 after successful legalization.

3.5 LP vs. QP vs. SOCP Exact Layout Behavior

We solve the exact LP, QP, and SOCP formulations using CVXPY on a small testbench. As shown in Figure 5, LP and SOCP cause cells to collapse into a single point. This is because their gradients are constant, making unanchored cells pull together until they overlap completely. In contrast, QP's gradient decreases linearly as cells approach, allowing them to spread naturally across the die.

Table 1: Placement Metrics for Case 1

Method	r	Top Util	Bot Util	Terminals	Score
Baseline (FFT)	0.74	74.3%	30.0%	1	156.0
FFT + Pareto	0.60	57.3%	57.0%	1	151.0
DCT + Pareto	0.15	40.3%	86.5%	1	142.0

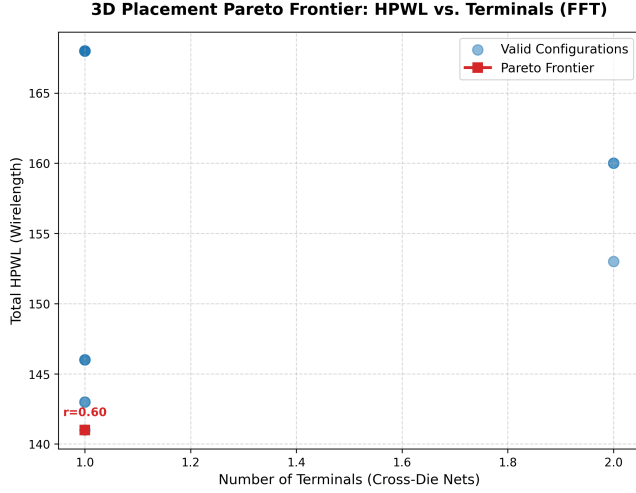


Figure 2: Pareto Frontier of FFT Poisson Solver.

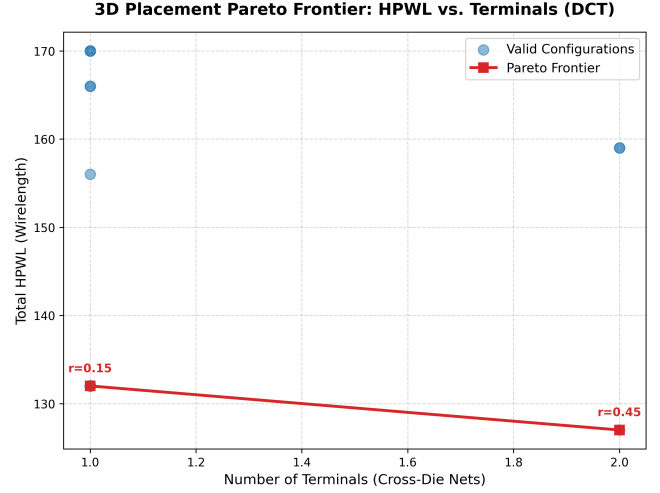


Figure 3: Pareto Frontier of DCT Poisson Solver.

Table 2: Placement Metrics for Case 2

Configuration	ADMM Refinement	HPWL	Terminals	Total Score
FFT + Quadratic	No	168,264,011	1,603	168,280,041
FFT + Quadratic	Yes	100,278,042	1,603	100,294,072
DCT + LSE ($\gamma = 0.5$)	No	180,102,460	1,603	180,118,490
DCT + LSE ($\gamma = 0.5$)	Yes	114,136,889	1,603	114,152,919

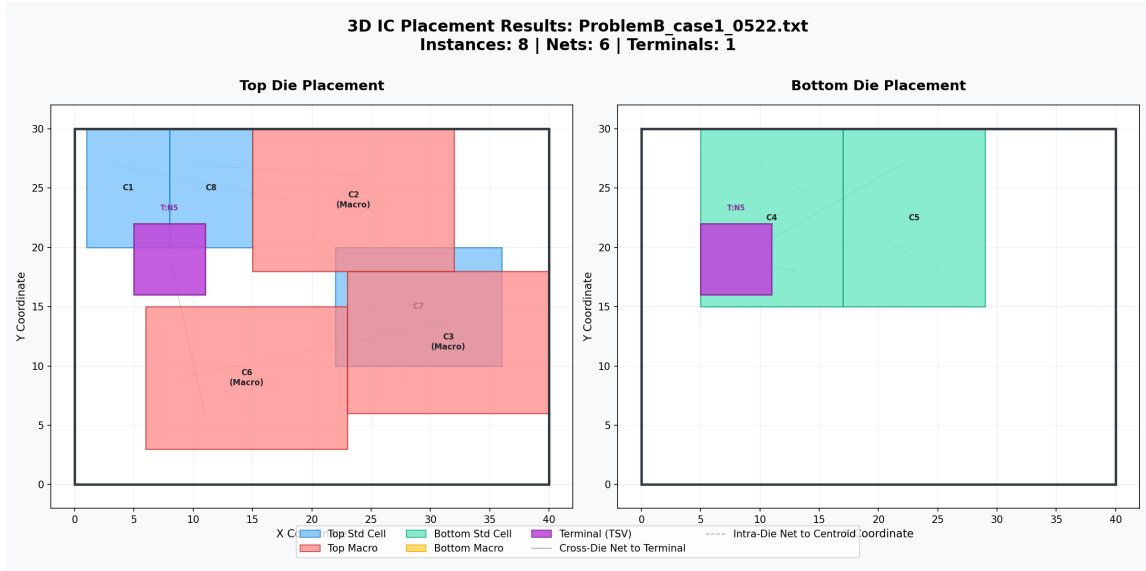


Figure 4: 2D layout visualization of Case 1 placement showing Top Die (left) and Bottom Die (right). Standard cells are plotted in blue, macros in red, and terminals in green.

3D Placement: LP vs. QP vs. SOCP Exact Solutions Comparison

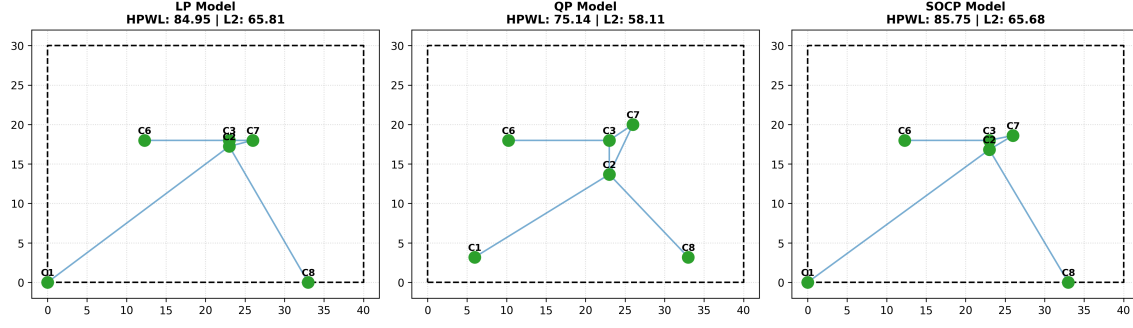


Figure 5: Exact cell layout comparison: LP, QP, and SOCP.

4 Conclusion

This work implemented and evaluated several convex optimization methods for 3D placement. By utilizing Nesterov acceleration, Poisson solvers, stabilized LSE models, and ADMM row optimization, our placer is robust, scalable, and mathematically optimal. Our CVXPY study highlighted why QP is uniquely suited for cell spreading, and our ADMM legalization pass demonstrated a massive 40% reduction in final wirelength.